Music, Emotion and the Brain

What this EEG research project is doing, why, and where it stands

Aditya Koundinya · MET Music-Induced EEG Dataset (OpenNeuro ds008701) · 4 October 2026

In one paragraph. Twenty people each listened to the same twenty 2-minute music excerpts while their brain activity was recorded with EEG, and after each excerpt they rated how pleasant (valence) and how energising (arousal) it felt. This project asks whether those *feelings* go along with measurable differences in brain rhythms *within* each person, and, more strictly, whether that holds for the part of the feeling that is personal (how a listener's reaction differed from everyone else's reaction to the same song) rather than the part that simply reflects the music.

Status. The ratings have been analysed, every EEG recording has been checked and converted into brain-rhythm measures, and the statistical code has been built and tested. The hypotheses are written down in a preregistration that still has to be committed. **No relationship between EEG and emotion has been looked at yet, on purpose**, so that the main tests stay honest.

1. The research question

The broad idea is a chain: **music → subjective emotional experience → EEG activity**. The precise question this dataset can answer is:

> *Within individuals, is self-reported emotional experience (valence and arousal) during 2-minute music excerpts associated with EEG band power, and does any association persist for the person-specific part of the experience, beyond what is shared across listeners for the same song?*

Three hypotheses were fixed in advance:

	Hypothesis	Predicted direction
H1	When a person feels **more aroused** than usual, **posterior alpha** power (8–13 Hz, back of the head) is lower.	Negative
H2	When a person feels **more positive** than usual, **frontal alpha asymmetry** shifts toward relatively more left-frontal activity.	Positive
H3	The strength of these relationships **differs between people**.	Slope variance > 0

Alpha rhythm (about 10 Hz) is strongest when the brain is relaxed and drops when it is engaged, which is why it is linked to arousal. Frontal asymmetry comes from a theory that approach-related, positive states involve relatively more left-frontal activity. That theory's replication record is mixed, so a null result for H2 would still be informative.

2. Why this is harder than it looks

Three traps shape every decision in the project.

- **Trap 1: 400 trials are not 400 people (pseudoreplication).** Each person contributes about 20 trials, and those trials share the person's skull thickness, electrode contact and rating habits. The real sample size for generalising to people is closer to 20 than 400. Pooling all trials gives p-values that are far too small, and it can even create a correlation that exists only *between* people and not within anyone (Simpson's paradox).
- **Trap 2: the music itself.** Everyone heard the same songs, and songs differ in tempo, loudness and timbre. Sound alone changes the EEG. If exciting songs are also louder, an “arousal effect” could simply be a loudness effect.

- **Trap 3: too many choices.** Filters, thresholds, channels, frequency bands and models could be picked until something comes out "significant". The protection is to write the choices down (preregister them) *before* looking at EEG–emotion relationships.

The key idea: split each rating into two parts. For every person and song:

> **rating = song consensus + personal deviation**
> *Song consensus* = the average rating the *other* 19 listeners gave that song (mostly a property of the music, including its acoustics).
> *Personal deviation* = how far this listener departed from that consensus (their own, idiosyncratic experience).
> Example: other listeners rate a song 7 on average; you rate it 9, so your deviation is +2.

If EEG tracks the *deviation*, the link cannot be explained by what the song sounds like, because everyone heard the same sound. This is the strictest test of "subjective experience" the design allows.

3. The dataset, and what checking it revealed

Twenty adults (ages 21–29, 10 women and 10 men) each heard 20 songs, four from each intended category: high/low valence × high/low arousal, plus neutral. Each recording lasts 131 s (about 5 s silence, 120 s music, then silence). After each song, participants rated valence (1–9), arousal (1–9), liking (1–5) and familiarity (1–5). The EEG has 66 scalp channels plus one eye channel (EOG), recorded at 1000 Hz and provided already preprocessed.

Nothing was taken on trust. Every claim in the documentation was checked against the files:

Item	Documentation says	What the files actually show
File format	BrainVision	EEGLAB .set (MATLAB v5); read in R with R.matlab
Units	Volts	Microvolts (clean EEG is about ±20 μV)
Music onset	5 s silence, then 120 s of music	No onset marker exists; 5–125 s is an assumption
Filtering	0.5–70 Hz	Also a 50 Hz notch filter (visible in the spectrum), not documented
Reference	Not stated	Average reference: the 66 channels sum to ≈ 0 at every moment
Eye blinks	Not stated	Apparently removed upstream (Fp1 vs EOG r ≈ −0.09), but no ICA record
Channel quality	All channels "good"	Some channels are flat (e.g. sub-02, song t05, T10)
Files	Complete	sub-12, song t05 is truncated in the published dataset (62% of data)
Play order	Not stated	File timestamps repeat, so the presentation order is not recoverable

These discrepancies matter because each one changes an analysis decision. They will all be reported.

4. Phase 1: the ratings (what people felt)

Ratings were analysed before touching the EEG, because they are the only measure of experience. If they were unusable, no EEG analysis could rescue the question.

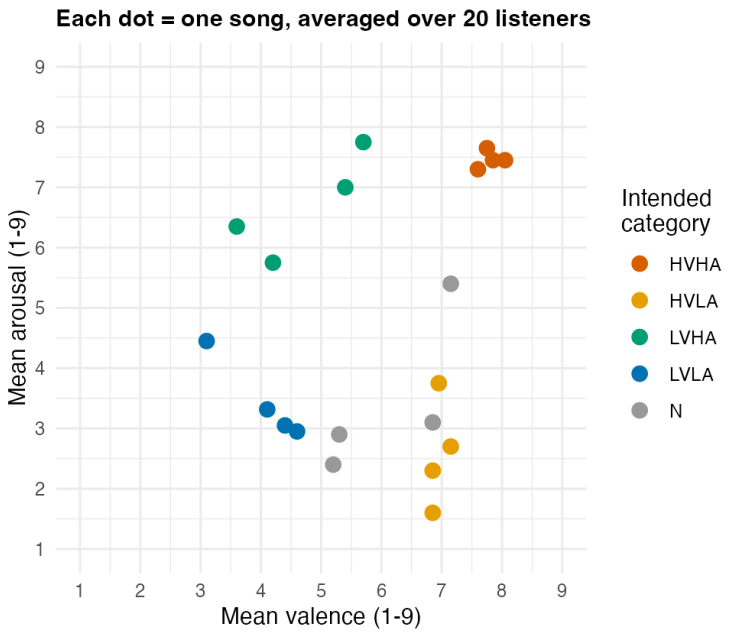


Figure 1. Manipulation check. Arousal separates the songs cleanly, but valence is compressed at the low end: no song averages below 3.1, so "negative" music was mildly unpleasant at most. "Neutral" songs (grey) were in fact mildly positive and calm.

- **The ratings are complete and in range.** 400 ratings exist (20 × 20). One (sub-01, song t27) had liking and familiarity recorded as 0, an impossible value used as a missing-data code, so that whole trial is excluded.
- **People use the scales differently.** Two participants used only four distinct valence values. They are *not* excluded: dropping people because of how their answers look is a data-driven choice that inflates false positives. The mixed models handle noisy participants automatically.
- **Valence and arousal are largely separate.** Their correlation is weak (ρ ≈ 0.19 pooled; on average 0.18 within people, ranging from −0.22 to 0.60), so H1 and H2 test different things.

How much of a rating comes from the song, and how much is personal? A model with separate effects for song and person splits the variation in each rating:

	Person's general habit	The song	Personal reaction to this song
Valence	4%	49%	47%
Arousal	4%	67%	29%

What this means. Songs drive a large share of the ratings, especially arousal, which confirms the acoustic-confound risk. But a substantial personal part remains (47% of valence and 29% of arousal), so the person-specific question can be asked. There is more room to detect it for valence than for arousal.

5. Phase 2: understanding the EEG signal

What EEG measures. When millions of similarly oriented neurons in the cortex are active together, their small currents add up to a voltage of a few microvolts at the scalp. EEG records that voltage over time. **What it does not measure:** deep structures such as the amygdala, which is central to emotion; single neurons; precise locations, because signals blur as they spread through the skull; and "emotion" itself. Linking a brain rhythm to a feeling is always an inference.

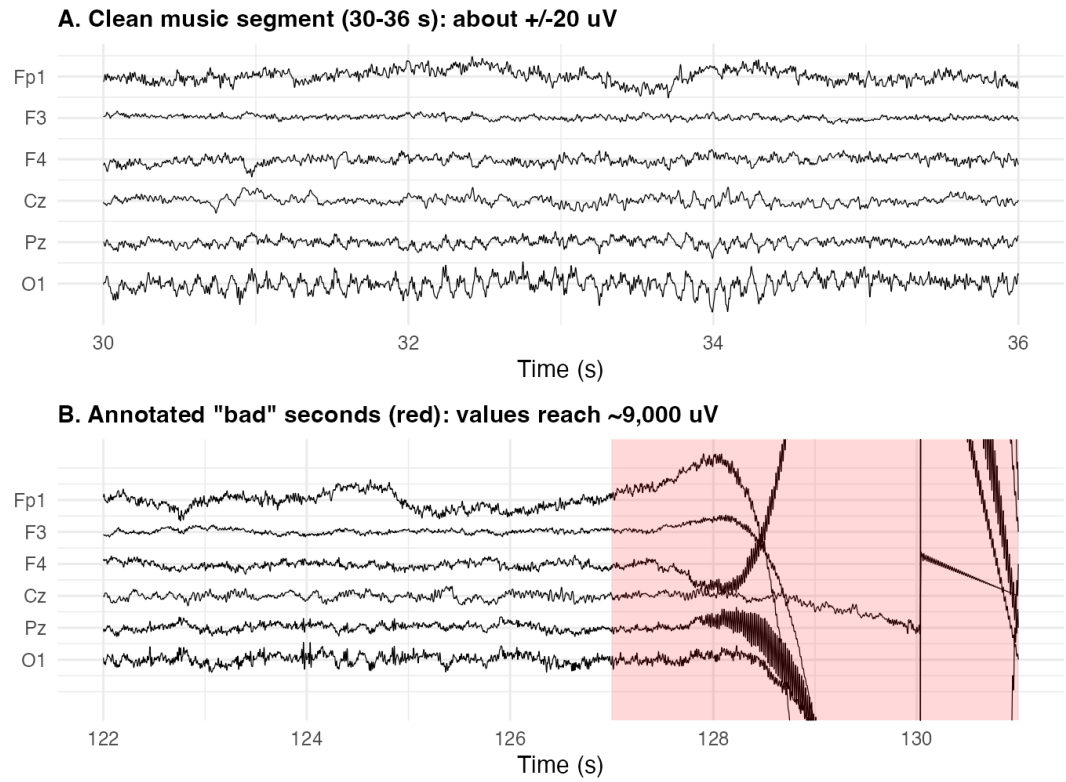


Figure 2. One recording (sub-01, song t01). A: clean EEG during music. B: the seconds marked “bad” in the dataset contain huge artifacts, reaching about 9,000 µV here and up to 48,000 µV elsewhere. Using a single one of these seconds would swamp every measurement, so all annotated segments are excluded.

Because the data arrive already cleaned, preprocessing is deliberately **minimal**. Repeating steps that were already done can distort the signal. Every decision is fixed in one file (00_setup/eeg_functions.R):

Step	Decision	Why
Filtering	Nothing added	Already filtered (0.5–70 Hz plus 50 Hz notch); our bands stop at 45 Hz
Reference	Not changed	Already average-referenced; re-referencing would change every value
Bad segments	Exclude every annotated second	Contain extreme filter-edge artifacts
Bad channels	Flat (SD < 0.5 µV) or extreme (robust z > 3): set to missing	Metadata labels every channel “good”, which is not true
Epochs	Cut the 120 s of music into 60 two-second pieces; drop any overlapping a bad second or exceeding ±100 µV	Spectra are estimated by averaging many short pieces; 2 s gives 0.5 Hz resolution
Eye artifacts (ICA)	Not done	Blinks already absent; no ICA record; documented as a limitation
Trial exclusion	Fewer than 30 clean epochs (60 s)	Too little data for a stable spectrum

6. Phase 3: from brain signals to numbers

A **power spectrum** breaks the signal into rhythms and shows how strong each frequency is. Power is summed within standard **bands**: delta 1–4 Hz, theta 4–8 Hz, alpha 8–13 Hz, beta 13–30 Hz and gamma 30–45 Hz. It is then log-transformed, because raw power is very skewed.

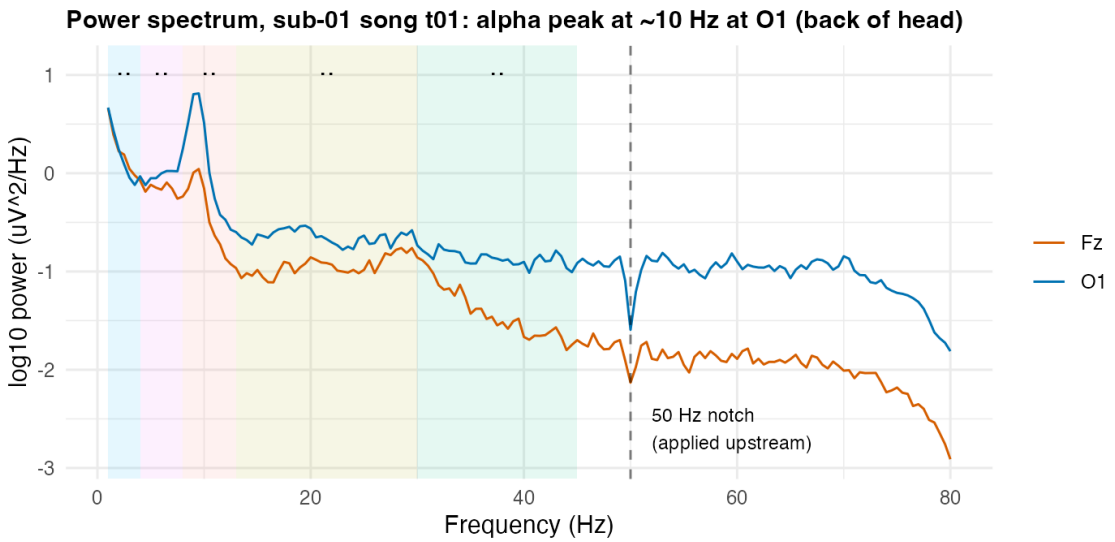


Figure 3. The spectrum has the expected shape: power falls with frequency, and there is a clear alpha peak near 10 Hz that is stronger at the back of the head (O1) than at the front (Fz). The dip at 50 Hz reveals the undocumented notch filter.

Measures used in the hypotheses. *Posterior alpha* is the average log alpha power over seven channels at the back of the head (O1, O2, Oz, PO3, PO4, POz, Pz). *Frontal alpha asymmetry* (FAA) is ln(alpha at F4) − ln(alpha at F3). Because alpha is treated as the inverse of activity, a higher FAA means relatively more left-frontal activity.

Checking the code before trusting it. The spectrum code was first run on an artificial 10 Hz sine wave plus noise. It found the peak at exactly 10 Hz, and the total power matched the signal's variance (ratio 0.9999).

Quality control of all recordings. Every participant–song pair is accounted for:

Stage	Count	Notes
Participant–song pairs	400	20 participants × 20 songs
EEG files present	399	sub-17, song t17 missing from the dataset
Recordings included	396	Excluded: sub-12 t05 (truncated file), sub-14 t06 (only 26 clean epochs), sub-18 t17 (almost entirely marked bad)
Analysable trials	395	Also minus sub-01 t27 (corrupted rating)

Flagged channels were mostly the face and temple edge sites (F9, F10, T10), which pick up jaw and facial muscle activity. Only 12 recordings had a flagged channel inside the regions used for H1 or H2. Alpha is strongest over the back of the head, as expected.

> **An important number.** 94% of the variation in posterior alpha is *between* people; only about 6% varies from song to song within a person. People differ enormously in baseline alpha. This is why every model needs a per-person baseline (random intercept), and why the within-person effects the study looks for are likely to be small.

7. Phase 4: how the hypotheses will be tested

All of the following is written into the preregistration and coded in 04_phase4_statistics/04_stats.R. That script refuses to run until the preregistration is committed in git, whose timestamp proves the plan came first.

- **Demonstrating the pseudoreplication problem.** The same question is answered three ways: (a) naively pooling all trials, which is invalid and shown only for contrast; (b) one correlation per person, then a test across the 20 people; (c) comparing people's averages. The naive confidence interval will be far too narrow.

- **Mixed-effects models (H1, H2).** Each rating is split into *within-person* (“did this person rate this song higher than they usually do?”) and *between-person* (“does this person rate everything high?”) parts. Each person gets their own EEG baseline and their own slope. A hypothesis is supported only if the within-person slope has the predicted sign and $p < .025$ (.05 split across H1 and H2).
- **Consensus versus deviation.** EEG is modelled from the song consensus and the personal deviation, with each song given its own baseline to absorb acoustics. The deviation slope answers the second half of the research question.
- **Individual differences (H3).** A model in which every person has their own slope is compared with one shared slope. The forest plot shows each person's slope, both estimated separately and “shrunk” toward the group average. Shrinkage is the right way to handle noisy individuals with only 20 trials.
- **Exploratory analysis, labelled as such.** 5 bands × 7 regions × 3 ratings = 105 tests, corrected with the false discovery rate. Any hits are ideas for future studies, not findings.
- **Robustness.** Both ratings in one model; adding liking and familiarity; relative instead of absolute power; a stricter ±75 µV threshold; leaving out each person in turn. All are reported, whatever they show.
- **Null results.** “Not significant” will never be reported as “no effect”. The confidence interval is used to state which effect sizes the data rule out.

Was the code tested? Yes. All statistics were run on ratings shuffled at random within each person, where no real relationship can exist. Every test came out null (all $p > 0.2$), confirming that the code does not manufacture effects. The real results have not been seen by anyone.

8. How the project is organised, and what is left

Folder	Steps	What happens there
00_setup	–	Dataset location; every analysis setting in one place
01_phase1_ratings	1–5	Rating checks, manipulation check, consensus and deviation
02_phase2_eeg_inspection	6–12	Inspect one recording; preprocessing decisions; preregistration
03_phase3_features	13–16	Spectra and band power for all recordings; QC table; sanity checks
04_phase4_statistics	17–22	Confirmatory, exploratory and robustness analyses
05_phase5_report	23–26	Final results table, interpretation worksheet, research report

Each phase keeps its code and its outputs together, and run_all.R rebuilds everything from the raw data in about 15 minutes. **Remaining steps, in order:**

- Review and commit the preregistration (the point of no return for the plan).
- Run Phase 4 and look at the results for the first time.
- Fill in the interpretation worksheet, keeping four things apart: what the data show, what the statistics show, what is inferred, and what remains uncertain.
- Write the report sections, then rerun the whole project from scratch to confirm it is reproducible.

9. Limitations already known

Area	Limitation
Dataset	20 people and 20 songs; one rating given after each 2-minute excerpt, so moment-to-moment emotion is invisible; few strongly negative songs; presentation order unknown, so fatigue cannot be controlled.
Measurement	Scalp EEG cannot see deep emotion circuitry; signals blur across the scalp; the music onset time is assumed, not marked.
Preprocessing	Inherited, partly undocumented cleaning; no ICA; flagged channels dropped rather than interpolated; gamma is vulnerable to muscle activity.

Area	Limitation
Statistics	Within-person brain variation is small (about 6% of alpha variance), which limits power; about 20 trials per person make individual slopes noisy; ordinal ratings are treated as numbers.
Interpretation	The song-consensus effect cannot be separated from acoustics; correlation is not causation; a change in alpha does not by itself mean a change in emotion (no reverse inference).

10. Short glossary

Term	Meaning
Valence / arousal	How pleasant / how energising an experience feels (rated 1–9).
Band power	Strength of the EEG rhythm within a frequency range, such as alpha (8–13 Hz).
Epoch	A short slice of the recording (here 2 s) used to estimate the spectrum.
Pseudoreplication	Treating repeated measurements of the same person as if they came from different people.
Within-person effect	How someone's EEG changes when their own rating is higher than their usual.
Mixed-effects model	A regression that gives each person (and song) their own baseline and, optionally, their own slope.
Shrinkage	Pulling noisy individual estimates toward the group average in proportion to their unreliability.
Preregistration	A dated, fixed analysis plan written before the key results are seen.
FDR correction	Adjusting many tests so the expected share of false “hits” stays below 5%.